

Strings: Extra Practice (Part 3)

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```
library(tidyverse)
library(rvest)
library(httr)
```

On Your Own - Extra practice with strings and regular expressions

18. Produce an R string which the regular expression represented by "\\..\\.\\.." matches. In other words, find a string y below that produces a TRUE in `str_detect`.

```
true_string <- ".a.a.a"  
str_detect(true_string, "\\...\\.\\.\\.\\.\\.")
```

```
[1] TRUE
```

19. What words in `stringr::words` have the highest number of vowels? What words have the highest proportion of vowels? (Hint: what is the denominator?) Figure this out using the tidyverse and piping, starting with `as_tibble(words) |>`.

```
words_tbl <- as_tibble(words)

words_tbl
```

```
# A tibble: 980 x 1
  value
  <chr>
1 a
2 able
3 about
```

```

4 absolute
5 accept
6 account
7 achieve
8 across
9 act
10 active
# i 970 more rows

```

```

words_tbl |>
  mutate(
    num_vowels = str_count(value, "[aeiou]"),
  ) |>
  arrange(desc(num_vowels)) |>
  print(n = 10)

```

```

# A tibble: 980 x 2
  value      num_vowels
  <chr>      <int>
1 appropriate      5
2 associate        5
3 available        5
4 colleague        5
5 encourage        5
6 experience        5
7 individual        5
8 television        5
9 absolute         4
10 achieve         4
# i 970 more rows

```

```

words_tbl |>
  mutate(
    num_letters = str_count(value, "."),
    num_vowels = str_count(value, "[aeiou]"),
    prop_vowels = num_vowels / num_letters
  ) |>
  arrange(desc(prop_vowels)) |>
  print(n = 10)

```

```

# A tibble: 980 x 4

```

	value <chr>	num_letters <int>	num_vowels <int>	prop_vowels <dbl>
1	a	1	1	1
2	area	4	3	0.75
3	idea	4	3	0.75
4	age	3	2	0.667
5	ago	3	2	0.667
6	air	3	2	0.667
7	die	3	2	0.667
8	due	3	2	0.667
9	eat	3	2	0.667
10	europe	6	4	0.667

i 970 more rows

20. *Carefully* explain what the code below does, both line by line and in general terms.

```
temp <- str_replace_all(words, "^[A-Za-z])(.*)([a-z])$", "\\3\\2\\1")
as_tibble(words) |>
  semi_join(as_tibble(temp)) |>
  print(n = Inf)
```

Joining with `by = join\_by(value)`

```
# A tibble: 45 x 1
  value
  <chr>
1 a
2 america
3 area
4 dad
5 dead
6 deal
7 dear
8 depend
9 dog
10 educate
11 else
12 encourage
13 engine
14 europe
15 evidence
16 example
```

```
17 excuse
18 exercise
19 expense
20 experience
21 eye
22 god
23 health
24 high
25 knock
26 lead
27 level
28 local
29 nation
30 no
31 non
32 on
33 rather
34 read
35 refer
36 remember
37 serious
38 stairs
39 test
40 tonight
41 transport
42 treat
43 trust
44 window
45 yesterday
```

Line 1: Create vector temp from words - swap the first and last letters of each word
Line 2: Turn words vector into a tibble and pipe
Line 3: Semi-join temp as tibble - will include only rows from words that are also in temp
Line 4: Print all rows

We create a 1-column tibble of words that have the identical last and first letters. ## Coco and Metacritic

We will check out the Metacritic page for the 2017 movie Coco, scrape information from that page (we'll get into web scraping in a few weeks!), clean it up into a usable format, and answer some questions using strings and regular expressions.

```
library(polite)
coco <- "https://www.metacritic.com/movie/coco-2017/" |>
```

```

bow() |>
scrape()

top_reviews <-
  "https://www.metacritic.com/movie/coco-2017/critic-reviews/" |>
  bow() |>
  scrape()
top_reviews <- html_nodes(top_reviews, ".leading-normal")
top_reviews <- html_text(top_reviews)

user_reviews <-
  "https://www.metacritic.com/movie/coco-2017/user-reviews" |>
  bow() |>
  scrape()
user_reviews <- html_nodes(user_reviews, ".leading-normal")
user_reviews <- html_text(user_reviews)

```

21. `top_reviews` is a character vector containing the 10 most recent critic reviews for Coco, while `user_reviews` is a character vector with the 50 most recent user reviews (although many are truncated).

- a) Print out the user reviews where the reviewer mentions “emotion” or “cry”. Think about various forms (“cried”, “emotional”, etc.) You may want to turn reviews to all lower case before searching for matches.

```

user_reviews |>
  str_to_lower() |>
  str_subset("\\bemotion|cry(ing)?|crie[sd]")

```

```

[1] "a disney classic - probably one of their best work just because of how authentic it is
[2] "one of pixar's top 5 movies. what makes this movie so good is not only an emotional s
[3] "yes yes yes, that's the good film. i cried so hard. good plot, music, characters, love
[4] "i honestly cannot believe how overrated this film is. it has higher ratings than many o
[5] "pixar explores mexican culture, probing the spirituality and mythos of day of the dead
dropping, as it constantly one-ups itself right alongside the developing plot. particularly s
[6] "\"coco.\" released in 2017, this animated gem takes us on a musical journey through mex
inspiring. it's a visual fiesta that celebrates the richness of mexican folklore and traditi
stirring serenade. the original songs, composed by michael giacchino and kristen anderson-
lopez, capture the essence of mexican music with infectious energy. the standout track, \"re
handed sentimentality and instead delivering a message that feels genuine and profound.\n\nni
[7] "i went to see this movie in theatres with my grandfather and walked out completely stun
see for all ages, and as a family if possible."

```

```
[8] "coco has some really great emotion and atmosphere in it. it's pixar's only good modern
[9] "a truly stunning movie. the animation in the land of the dead made me breathless. i fi
[10] "best and unique story line movie from pixar animation studio. this movie was so emotion
[11] "i love this movie. coco is easily pixar's best movie since inside out, it's not even c
```

- b) In critic reviews, replace all instances where “Pixar” is used with its full name: “Pixar Animation Studios”.

```
top_reviews |>
  str_replace_all("Pixar|pixar", "Pixar Animation Studios")
```

```
[1] "At every imaginative juncture, the filmmakers (the screenplay is credited to Pixar Ani
[2] "Thankfully, Coco, Pixar Animation Studios’s latest original work and one of their very
[3] "Coco is the best-looking Pixar Animation Studios movie since the tonally uneven "The G
[4] "Coco is packed with terrific original tunes such as "Remember Me" (by Robert Lopez and
Lopez of "Frozen") and "Proud Corazón" (co-written by Adrian Molina, the film’s co-
director). But it’s not your average musical, in which characters wail their wants and feelin
[5] "I had some minor quibbles about Coco while I was watching it, but I can’t remember wha
[6] "It’s heart that’s overflowing with love, poignancy, humor, color and music."
[7] "Pixar Animation Studios has raised the animation bar again, with its most musical - an
[8] "If an animated movie is going to offer children a way to process death, it’s hard to e
[9] "This is a masterpiece of woven ancestral roots and the importance of "familia," confide
[10] "Pixar Animation Studios’s latest animated film may lack the volume of out-
loud laughs of the "Toy Story" series, but the fantasy set in Mexico doesn’t skimp on the tea
hour feature film."
```

- c) Find out how many times each user uses “I” in their review. Remember that it could be used as upper or lower case, at the beginning, middle, or end of a sentence, etc.

```
i_tbl = as_tibble(user_reviews) |>
  mutate(num_i = str_count(str_to_lower(value), "\\bi\\b"))

i_tbl
```

A tibble: 50 x 2

	value	num_i
	<chr>	<int>
1	"hello data science 2 class. this review needs to be at least 75 chara~	0
2	"A Tear-Stained Celebration of Overflowing Love, Mexican Heritage, and~	0
3	"Que obra perfeita, Sinceramente,se tornou meu filme favorito de anima~	0
4	"a disney classic - probably one of their best work just because of ho~	0
5	"[SPOILER ALERT: This review contains spoilers.]"	0

```

6 "Best animated movie I've watched until now. It has literally shaped m~      1
7 "One of Pixar's top 5 movies. What makes this movie so good is not onl~      0
8 "[SPOILER ALERT: This review contains spoilers.]"                          0
9 "7.5/10\n\n0 maior ponto positivo desse filme é a originalidade tanto ~      0
10 "Viva: A Vida é uma Festa não é só um dos melhores filmes da Pixar, é ~      0
# i 40 more rows

```

- d) Do critics or users have more complex reviews, as measured by average number of commas used? Be sure your code weeds out commas used in numbers, such as “12,345”.

```

users_cnt <- as_tibble(user_reviews) |>
  mutate(num_coms = str_count(value, "(?<!\d),|,(?!\\d)"))

users_cnt

```

```

# A tibble: 50 x 2
  value                                num_coms
  <chr>                                <int>
1 "hello data science 2 class. this review needs to be at least 75 ch~      0
2 "A Tear-Stained Celebration of Overflowing Love, Mexican Heritage, ~      8
3 "Que obra perfeita, Sinceramente,se tornou meu filme favorito de an~      3
4 "a disney classic - probably one of their best work just because of~      0
5 "[SPOILER ALERT: This review contains spoilers.]"                          0
6 "Best animated movie I've watched until now. It has literally shape~      0
7 "One of Pixar's top 5 movies. What makes this movie so good is not ~      1
8 "[SPOILER ALERT: This review contains spoilers.]"                          0
9 "7.5/10\n\n0 maior ponto positivo desse filme é a originalidade tan~      4
10 "Viva: A Vida é uma Festa não é só um dos melhores filmes da Pixar,~      5
# i 40 more rows

```

```

critics_cnt <- as_tibble(top_reviews) |>
  mutate(num_coms = str_count(value, "(?<!\d),|,(?!\\d)"))

critics_cnt

```

```

# A tibble: 10 x 2
  value                                num_coms
  <chr>                                <int>
1 At every imaginative juncture, the filmmakers (the screenplay is cr~      5
2 Thankfully, Coco, Pixar's latest original work and one of their ver~      4
3 Coco is the best-looking Pixar movie since the tonally uneven "The ~      1

```

4	Coco is packed with terrific original tunes such as "Remember Me" (~	2
5	I had some minor quibbles about Coco while I was watching it, but I~	1
6	It's heart that's overflowing with love, poignancy, humor, color an~	3
7	Pixar has raised the animation bar again, with its most musical - a~	2
8	If an animated movie is going to offer children a way to process de~	2
9	This is a masterpiece of woven ancestral roots and the importance o~	1
10	Pixar's latest animated film may lack the volume of out-loud laughs~	1

```
mean(users_cnt$num_coms)
```

```
[1] 3.34
```

```
mean(critics_cnt$num_coms)
```

```
[1] 2.2
```

On average, users have a higher average number of commas used. This is an interesting finding and suggests that users have more complex reviews (by this metric).